

1. Describe two important environment contrasts that are generally considered for designing rational agents. (2)
 2. Prove or disprove $\exists x(\text{Mat}(x) \wedge \text{Tk}(\text{Hab}, x))$ from the following knowledgebase using resolution. (3)
 - a) $\text{Std}(\text{Hab})$
 - b) $\text{Mat}(\text{MatFor}(y)) \vee \neg \text{Std}(y)$
 - c) $\neg \text{Std}(y) \vee \text{Tk}(y, \text{MatFor}(y))$
 3. Elaborate the concept of Unification? (2)
 4. Show that
$$\forall x, y, z ((\text{Br}(x, y) \wedge \text{Sn}(z, x)) \Rightarrow \text{Nw}(z, y))$$
is a syntactically correct sentence of First Order Logic. (3)
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1. Explain the characteristics of a rational Agent. (2)
 2. Convert the following FOL sentence into CNF. (3)
$$\forall x (\text{Std}(x) \wedge \text{Tk}(x, \text{CSE1203}) \leftrightarrow \text{CStd}(x))$$
 3. Why backward chaining algorithms are said to be goal oriented? (2)
 4. Prove or disprove the proposition M using a typical forward chaining algorithm on the knowledgebase that follows. (3)
$$\begin{aligned} &A, B, C, D, G \wedge M \Rightarrow H, A \wedge H \Rightarrow E, C \wedge K \Rightarrow L, H \wedge K \Rightarrow S, B \wedge F \Rightarrow G, B \wedge D \Rightarrow K, \\ &D \wedge F \Rightarrow G, D \wedge G \Rightarrow M, D \wedge L \Rightarrow H. \end{aligned}$$
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1. How is a Term defined in the syntax of First Order Logic? (2)
 2. Resolve $P2(\text{"Bt"}, \text{"Rt"})$ from the following clauses, considering that the variables are standardized. (3)
$$\begin{aligned} &P1(y, F1(x)) \vee P2(y, u) \\ &\neg P1(\text{"Bt"}, z) \vee P3(u, F2(x)) \\ &\neg P3(\text{"Rt"}, v) \end{aligned}$$
 3. Explain the matching rules that are involved during Unification. (2)
 4. Derive the most general unifier, if possible, for the following set of FOL sentences. (3)
$$S = \{P1(F2(\text{"Mk"}), z, \text{"Bl"}, u), P1(y, \text{"Rd"}, x, F3(z))\}$$
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1. What are the basic components of an Agent? (2)
2. Consider the following situation of 'Find Gold avoiding Monsters' problem.
$$\begin{aligned} S_{i,j} &\Leftrightarrow M_{i-1,j} \vee M_{i+1,j} \vee M_{i,j-1} \vee M_{i,j+1} & (i \geq 1) \wedge (j \geq 1) \wedge (i \leq 4) \wedge (j \leq 4) \\ \neg M_{1,1} \wedge \neg M_{2,1} \wedge S_{2,1} \wedge \neg G_{2,1} & A_{2,1} \end{aligned}$$
Derive $M_{3,1} \vee M_{2,2}$ using the inference rules of propositional logic. (3)
3. How Resolution is used to prove or disprove a sentence? (2)
4. Convert the following English sentence to CNF of FOL. (3)

A student of the department of EEE must take EEE1101 and PHY1201.